

Homework 3

Max Points: 50

Directions: Answer each of the questions to the best of your ability. Make sure to show your work in order to receive potential FULL credit. Just stating the answer will result in point deductions (and possibly no credit for the respective problem) and that cannot be disputed through the grade dispute policy. Your answers must typed up and submitted as a PDF. Handwritten solutions will not be accepted.

1. (10 points) Construct a context-free grammar for $L = \{a^n b^m c^k, n \geq 0, m \geq 0, k \geq 0, k = |n - m|\}$.
2. (15 points) Convert the following grammar into Chomsky Normal Form. In this problem, you must list your steps as you convert the grammar in order to be considered for partial credit (including serious attempt points).

$$\begin{aligned} S &\rightarrow AB \mid aB \mid C \mid \lambda \\ A &\rightarrow aA \mid a \mid B \mid \lambda \\ B &\rightarrow bB \mid b \mid C \mid A \\ C &\rightarrow AB \mid b \mid \lambda \end{aligned}$$

3. (5 points) Find an s-grammar that generates all strings from the following language definition: $L = \{a^n b^n : n \geq 2\}$
4. Given the following grammar:

$$\begin{aligned} S &\rightarrow AB \mid C \\ A &\rightarrow aAb \mid ab \\ B &\rightarrow cBd \mid cd \\ C &\rightarrow aCd \mid aDd \\ D &\rightarrow bDc \mid bc \end{aligned}$$

- (a) (5 points) Draw a derivation tree (parse tree) for the string $aabbccdd$.
 - (b) (5 points) Is the above grammar ambiguous? Yes or No. Based on your answer, you must prove it.
5. (10 points) Construct an NPDA (using a state diagram) that accepts the language $L = \{w : n_a(w) < n_b(w)\}$ where $\Sigma = \{a, b, c\}$. Please also specify your stack alphabet in Γ . Please also use the same notation that is used in class when labeling your transitions. If different notations are used, then no credit will be given for the question. This includes not receiving the serious attempt points. If a state diagram is not used, then no credit will be given as well.